

Mohammad Elious Ali Mondal

Date of Birth	12 December 1998	Website	EliousMondal.github.io
Mobile Phone	+1 5857541233	Github	github.com/EliousMondal
Email	mmondal@ur.rochester.edu	Google Scholar	M. Elious Mondal

Education

Ph.D.

2021-2026 Ph.D. in Chemistry at University of Rochester

Masters

2021-2023 Masters in Chemistry at University of Rochester

Bachelors and Masters

2016-2021 Integrated BS-MS from Indian Institute of Science Education and Research, Bhopal
Major - Chemistry, CGPA - 8.68/10

Publications

1. *Symmetry Adapted Hierarchical Equations of Motion for Exact Simulations of Large Polariton Systems*
M. E. Mondal, P. Huo. - *arXiv* (2026)
2. *A macroscopic condensation theory for vibrational strong coupling effects*
M. E. Mondal, S. M. Vega, P. Huo. - *Nat Commun* **17**, 8592 (2026)
3. *Long-Lived Polaritonic Coherence and Polaron Decoupling Effects in 2D Electronic Spectra.*
M. E. Mondal, S. Park, M. Son, A. N. Vamivakas, S. T. Cundiff, T. D. Krauss, P. Huo. - *PNAS* **123**, 17 (2026)
4. *Ab Initio Polariton Transport Dynamics with the Classical Path Approximation*
B. Chng, B. M. Weight, M. E. Mondal, P. Huo. - *Nano Letters* **26**, 18, 6055–6062 (2026)
5. *Collective Effects in Polariton Chemistry and Photophysics*
W. Ying, M. E. Mondal, E. R. Koessler, S. M. Vega, P. Huo. - *Ann. Rev. Phys. Chem.* **77**, 201-224 (2026)
6. *Polariton Spectra under the Collective Coupling Regime. II. Efficient Simulation of 2DES Spectra*
M. E. Mondal, A. N. Vamivakas, S. T. Cundiff, T. D. Krauss, P. Huo. - *J. Chem. Phys.* **162** (7), 074110 (2025)
7. *Polariton Spectra under the Collective Coupling Regime. I. Efficient Simulation of Linear Spectra*
M. E. Mondal, A. N. Vamivakas, S. T. Cundiff, T. D. Krauss, P. Huo. - *J. Chem. Phys.*, **162**, 014114 (2025)
8. *Quantum Dynamics Simulations of the Polariton Transport*
B. Chng, M. E. Mondal, W. Ying, P. Huo. - *Nano Letters* **25** (4), 1617-1622 (2025)
9. *Theory and Quantum Dynamics Simulations of Exciton-Polariton Motional Narrowing*
W. Ying, M. E. Mondal, P. Huo. - *J. Chem. Phys.*, **161**, 064105 (2024)
10. *Universal Measure for the Impact of Adiabaticity on Quantum Transitions.*
R. Pant, P. K. Verma, C. Rangi, M. E. Mondal, M. Bhati, V. Srinivasan, S. Wuster. - *Phys. Rev. Lett.* **132** (12), 11 (2024)
11. *Quantum Dynamics Simulations of the 2D Spectroscopy of Exciton Polariton.*
M. E. Mondal, E. Koessler, A. N. Vamivakas, S. T. Cundiff, T. D. Krauss, Pengfei Huo. - *J. Chem. Phys.*, **159**, 094102 (2023)

Presentations and Talks

1. American Chemical Society Spring (ACS-Spring) 2025 - **Poster**
2. American Physical Society March Meetings (APS-March) 2025 - **Talk & Poster**
3. American Conference on Theoretical Chemistry (ACTC)-2024- **Poster**
4. Winter school on Quantum Information for Chemistry (2024) -**Poster & Talk**
5. Graduate Research Symposium 2024, University of Rochester - **Poster**
6. Gordon Research Conference - Light Matter Interactions (GRC) 2023 - **Poster**
7. Telluride School for Theoretical Chemistry 2023, Telluride Science - **Poster**
8. TDDFT Summer School 2023 - **Poster**

Teaching

1. **CHM 451 (Fall 2023):** Teaching Assistant for Quantum Chemistry (Graduate level course)
2. **Science Outreach (Summer 2023):** Journey to Molecular World - a short summer course for K-12 students aimed as a brief introduction to Theoretical and Computational Chemistry
3. **CHM 451 (Fall 2022):** Teaching Assistant for Quantum Chemistry (Graduate level course)
4. **CHM 251(/441) (Fall 2022):** Teaching Assistant for Physical Chemistry - I (Undergraduate level course)
5. **Science Outreach (Summer 2022):** Journey to Molecular World - a short summer course for K-12 students aimed as a brief introduction to Theoretical and Computational Chemistry
6. **CHM 132 (Spring 2022):** Teaching Assistant for General Chemistry - II laboratory
7. **CHM 131 (Fall 2021):** Teaching Assistant for General Chemistry - I laboratory (Undergraduate level course)

Computational Skills

1. **Programming Languages:** Python, C++, Cython, Julia, $\text{\LaTeX}$, Git
2. **Libraries & Tools:** Eigen, Numpy, Scipy, Numba, PyTorch, CuPy, QuTip, Pandas, Matplotlib, ASE
3. **Computing/HPC:** open-mpi, mpi4py, Command-line(Linux), Bash
4. **Softwares:** INQ, NWChem, Newton-X, Turbomole, Gaussian, Avogadro, PySCF

Awards an Achievements

1. Finalist of **ACS Physical Chemistry Graduate Awards (2025)** in Theoretical Chemistry
2. **Agnes M. and George Messersmith Fellowship (2025)** It is one of the University's most competitive dissertation fellowships and is given to students who display exceptional ability and promise.
3. **Wu Fellowship (2024)** by Department of Chemistry, University of Rochester. Selected by the faculty on the basis of outstanding research support in chemistry
4. **The Journal of Chemical Physics poster award** at the American Conference on Theoretical Chemistry, (ACTC-2024).
5. **Wu Fellowship (2023)** by Department of Chemistry, University of Rochester. Selected by the faculty on the basis of outstanding research support in chemistry
6. **Kishore Vaigyanik Protsahan Yojana (KVPY) fellowship (2016-2021)** by Department of Science and Technology (DST) - India, to pursue a career in Fundamental Sciences.